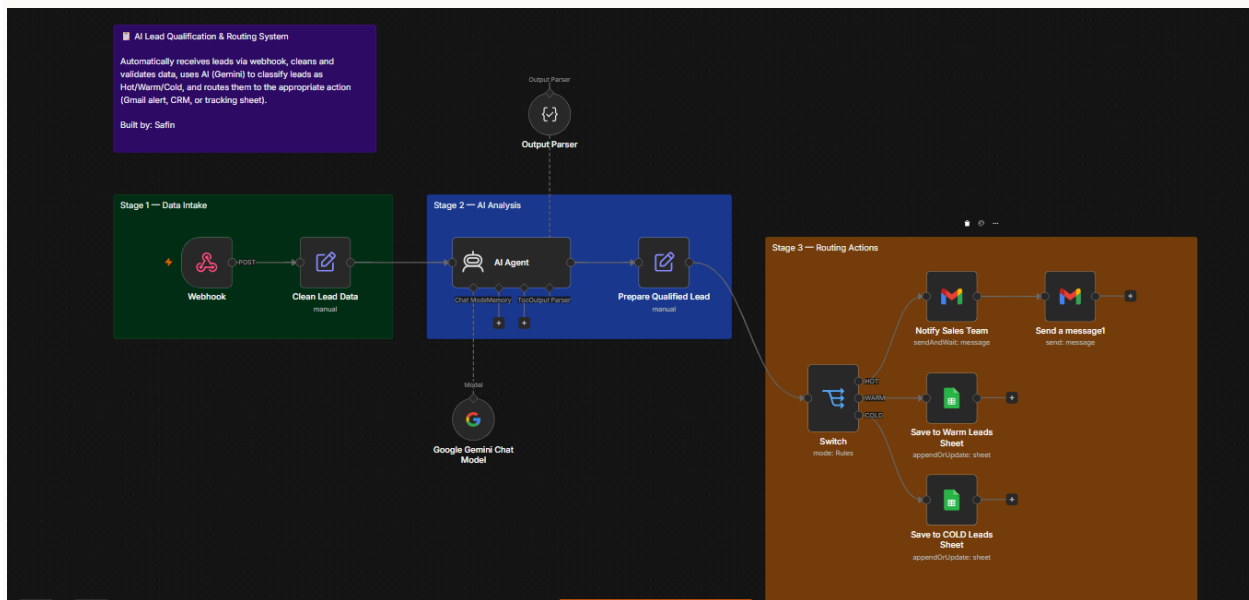
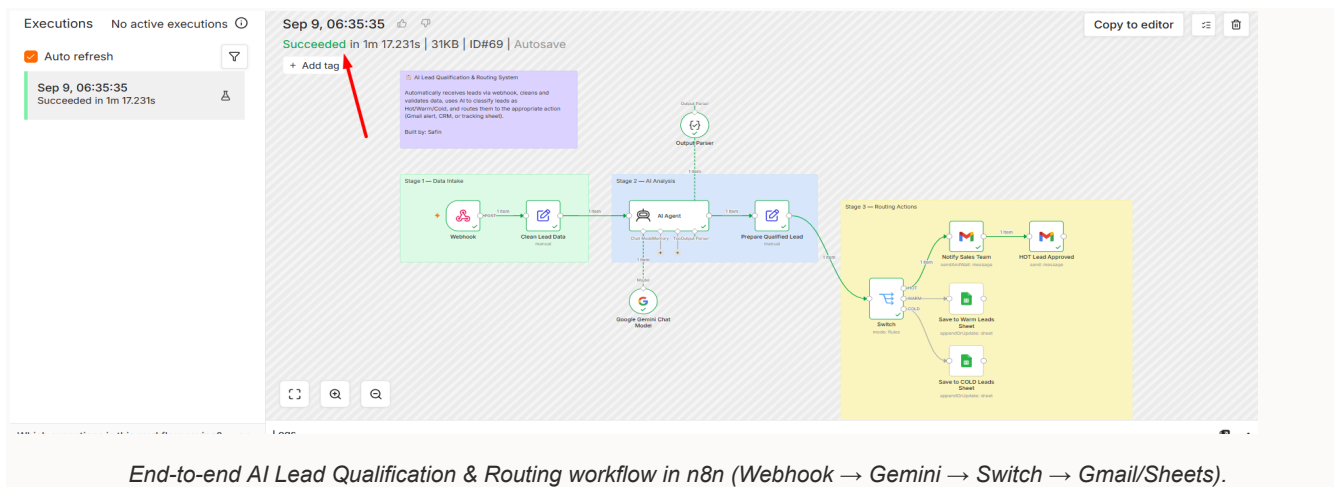


Case Study #1 — AI Lead Qualification & Routing System (n8n + Gemini)

Project Summary

This n8n automation receives inbound leads via Webhook, cleans/standardizes the data, and uses Google Gemini to qualify each lead as HOT / WARM / COLD (with a score + reason). Based on the AI result, the workflow automatically routes the lead to the correct destination—instant sales notification for HOT, and Google Sheets tracking for WARM/COLD.





The Problem

Many businesses receive leads from different sources.

The data is often inconsistent, and manually reviewing every lead takes time.

That can lead to:

- Slow response times
- Missed high-intent leads
- Too much manual work
- Poor lead prioritization

What I Built

I built an automated AI lead qualification system that:

- Receives leads via Webhook
- Cleans and standardizes the data
- Uses Gemini to qualify each lead
- Returns a structured result
- Routes the lead based on its status

Lead Routing

HOT → Gmail alert to sales

WARM → Google Sheets

COLD → Google Sheets

Actual Workflow Architecture

Stage 1 — Data Intake

Webhook → Clean Lead Data

Stage 2 — AI Analysis

AI Agent (Gemini Chat Model) → Output Parser → Prepare Qualified Lead

Stage 3 — Routing Actions

Switch (Rules) → HOT / WARM / COLD

- HOT → Notify Sales Team (Gmail) → HOT Lead Approved (Gmail message)
- WARM → Append to Warm Leads (Google Sheets)
- COLD → Append to Cold Leads (Google Sheets)

AI Qualification Logic (Prompt Rules Used)

Gemini is instructed to return ONLY valid JSON:

JSON

```
{
  "lead_status": "HOT",
  "lead_score": 85,
  "reason": "Short explanation",
  "priority": "HIGH",
  "recommended_action": "Contact immediately"
}
```

Classification rules:

- HOT: high budget, strong intent, clear need, ready to start soon
- WARM: genuine need, moderate intent, needs more info
- COLD: low intent/budget, unclear requirement

The screenshot shows the Gemini AI node configuration. The prompt is: "You are an expert sales lead qualification assistant. Analyze the lead information below. Classify the lead as: You are an expert sales lead qualification assistant. A...". The output is structured JSON:

```

{
  "lead_score": 95,
  "priority": "HIGH",
  "lead_status": "HOT",
  "reason": "The lead has a high budget ($50,000), a clear business need for AI Automation, strong buying intent, and is ready to start immediately this month.",
  "recommended_action": "Contact immediately"
}

```

Gemini AI returns structured JSON (status, score, reason, priority, action) for reliable routing.

Prepare Qualified Lead (Data Preparation)

This step maps AI output fields into the final lead object and keeps original lead fields (name/email/company/phone/budget/service/message), adding:

- lead_status
- lead_score
- priority
- reason
- recommended_action

Result: every downstream node receives a single standardized lead record.

The screenshot shows the 'Prepare Qualified Lead' node configuration. The 'Parameters' tab shows mappings for lead_status, lead_score, priority, and reason. The 'OUTPUT' tab shows the final standardized lead record:

```

{
  "name": "John Smith",
  "email": "john@example.com",
  "company": "Smith Digital Agency",
  "phone": "1555 123 4567",
  "budget": "50000",
  "service": "AI Automation",
  "message": "We want to start this month.",
  "lead_status": "HOT",
  "lead_score": 95,
  "priority": "HIGH",
  "reason": "The lead has a high budget ($50,000), a clear business need for AI Automation, strong buying intent, and is ready to start immediately this month.",
  "recommended_action": "Contact immediately"
}

```

Prepare Qualified Lead merges raw lead fields + AI qualification into a single standardized record.

Routing & Notifications

Once qualified, leads are routed automatically:

- HOT: sales team receives an immediate Gmail alert with lead details + AI reasoning
- WARM/COLD: saved to Google Sheets for follow-up and tracking

New HOT Lead

Name: John Smith
Company: Smith Digital Agency
Budget: 50000

Score: 95
Reason: The lead has a high budget (\$50,000), a clear business need (AI Automation), strong buying intent, and is ready to start this month.

Should we contact this lead?

Automated HOT lead notification sent to sales with score, reason, and recommended next action.

Expected Business Outcome

This is a portfolio project, not a real client engagement, so no real business metrics are claimed.

Expected outcomes:

- Faster lead triage
- Faster response to high-intent leads
- Less manual qualification
- Better lead organization
- More consistent prioritization

Business Use Cases

This workflow can be adapted for:

Real Estate

Qualify buyers, investors, and property inquiries.

Agencies

Prioritize new service inquiries.

SaaS

Score demo and sales leads.

B2B Sales

Identify high-value opportunities.

Recruitment

Pre-screen incoming candidates.

Tools & Stack

- Automation: n8n
 - AI: Google Gemini (Gemini Chat Model) via n8n AI Agent
 - Input: Webhook
 - Routing: Switch (Rules)
 - Notifications: Gmail
 - Tracking: Google Sheets
 - Output reliability: Structured Output Parser (JSON)
-